A CONDITIONAL SKEIN-LASAGNA OBSTRUCTION FOR A TRACE-73 CAPPELL–SHANESON CANDIDATE, PENDING GEOMETRIC IDENTIFICATION

ANONYMOUS REPOSITORY AUDIT

Abstract. We study a proposed trace-73 Cappell–Shaneson construction associated with the matrix

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

A public finite calculation produces the nonzero cubic Burau scalar -59072 in the grading convention corresponding to quantum degree 494. An exact Artin–Magnus certificate and the pure-braid Andreadakis theorem establish a third-order property of the public braid. We prove an abstract obstruction: if the compact word ledger is realized by the required framed Kirby presentation, if the Burau functional is identified with the specified rational $N = 2$ Manolescu–Walker–Wedrich coefficient trace, and if the recorded owner lifts realize the actual three-handle maps, then the candidate is a homotopy sphere not diffeomorphic to S^4. These three candidate-specific identifications remain open and are stated explicitly as hypotheses. The Lean development checks the finite and quotient algebra; it does not construct the external geometry or formalize four-dimensional differential topology.

Contents

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1. Introduction

The smooth four-dimensional Poincare conjecture asks whether every smooth closed manifold homotopy equivalent to S^4 is diffeomorphic to S^4. Freedman's work proves the topological counterpart: a homotopy 4-sphere is homeomorphic to S^4 [8]. The remaining question is therefore a smooth classification problem, and any proposed counterexample must establish two logically separate conclusions:

(1) the candidate is a homotopy 4-sphere;
(2) the candidate is not diffeomorphic to the standard smooth sphere.

Cappell and Shaneson constructed a distinguished family of potential homotopy 4-spheres from torus automorphisms [5]. Their handle structures were studied by Aitchison and Rubinstein [1]. Many subfamilies were subsequently shown to be standard, through work including Akbulut [2], Gompf [9], Kim–Yamada [12], and Iwaki [11]. This history makes geometric identification especially important: agreement of a matrix, a relator, or a trace parameter is not a substitute for identifying a complete framed handle presentation.

Skein lasagna modules extend Khovanov–Rozansky link homology to smooth four-manifolds [15]. Manolescu and Neithalath gave a two-handlebody description [13], and Manolescu, Walker, and Wedrich gave formulas compatible with general handle decompositions [14]. These results provide an attractive route to conclusion (ii): a nonzero class in a nonzero bidegree cannot be carried by a diffeomorphism to the lasagna module of S^4, which is concentrated in bidegree zero.

We investigate whether the public finite class can be constructed for the standard-form Cappell–Shaneson parameter $(41, 189, 73)$. Compact product-ribbon, point-push, owner-lift and Nielsen ledgers replace parts of the unavailable historical data, but they do not by themselves give an embedded framed Kirby equivalence, the candidate-specific quantum-trace map, or embedded attaching spheres with their actual MWW hemisphere maps. The paper therefore separates the conditional theorem from these comparison gaps.

Contributions. The main contributions are as follows.

(1) We give a compact word-level Aitchison–Rubinstein ledger and a six-sweep construction of the public point-push braid, while isolating the missing framed-embedding/Kirby equivalence.
(2) We identify the local checked R-matrix with the public Burau blocks and prove a full-entry third-order statement, while isolating the missing candidate-specific MWW-to-endpoint comparison and simultaneous transport.
(3) We give integral owner lifts, a Nielsen ledger, and the required core-counit algebra, while isolating the missing embedded sphere movies and their MWW coordinate maps.
(4) We provide executable ledgers, mutation tests and a kernel-checked Lean algebraic core.

Historical files referenced by earlier versions of the project are not publicly available. Their hashes are retained as provenance. The compact scripts regenerate useful projections of the proposed geometry but do not yet regenerate every load-bearing candidate-specific construction.

2. The trace-73 Cappell–Shaneson candidate

Let

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}. \tag{1}$$

This is Iwaki's standard form $X_{41,189,73}$; the triple occurs among the trace-73 representatives in his table [11, Table 2]. Direct integer arithmetic gives

$$\det A = 1, \qquad \det(A - I) = 1. \tag{2}$$

For $A \in \mathrm{SL}(3,\mathbb{Z})$, let $f_A : T^3 \to T^3$ be the induced diffeomorphism, made equal to the identity near a chosen point. Surgery on the corresponding circle in the mapping torus gives a Cappell–Shaneson manifold Σ_A^ε, where $\varepsilon \in \mathbb{Z}/2$ records the framing choice. The standard criterion says that Σ_A^ε is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$; see Iwaki's Proposition 2.1 and the original sources cited there [11, 5].

Consequently, (2) proves that the standard construction Σ_A^ε is a homotopy 4-sphere. It does not prove that a separately generated large Kirby diagram is a diagram of that construction. That identification is the first and most load-bearing hypothesis below.

Iwaki proves standardness for many infinite families. The appearance of $(41, 189, 73)$ in a representative table is not a standardness or exoticness theorem for this individual sphere. Likewise, the trace bound in Kim–Yamada's work stops below 73 [12]. These observations explain the choice of candidate but do not decide its diffeomorphism type.

3. Precise statements

We first state the abstract theorem in the form actually proved by the Lean development. This prevents geometric intuition from silently strengthening the result.

Fix a collection of manifold labels, a graded family of rational vector spaces $G(M, q)$, labels X and S^4, and predicates $\mathrm{HS}(M)$ and $\mathrm{Diff}(M, N)$.

Theorem 3.1 (Abstract nonstandardness theorem)**.** *Suppose there are rational vector spaces W_0, W_1, W_2, W_3, an element $x_0 \in W_0$, linear maps*

$$W_0 \xrightarrow{q_{01}} W_1 \xrightarrow{q_{12}} W_2, \qquad \ell_i : W_i \longrightarrow \mathbb{Q} \quad (i = 0, 1, 2),$$

and linear isomorphisms

$$T : W_2 \xrightarrow{\cong} W_3, \qquad F : W_3 \xrightarrow{\cong} G(X, 494),$$

such that

$$\ell_0(x_0) = -59072, \tag{3}$$

$$\ell_1 q_{01} = \ell_0, \tag{4}$$

$$\ell_2 q_{12} = \ell_1. \tag{5}$$

Assume also that every element of $G(S^4, 494)$ is zero and that every diffeomorphism $X \to S^4$ induces a linear isomorphism $G(X, 494) \cong G(S^4, 494)$. Then X is not diffeomorphic to S^4.

Proof. Set $v = F(T(q_{12}(q_{01}(x_0))))$. If $v = 0$, injectivity of F and T implies $q_{12}(q_{01}(x_0)) = 0$. Applying ℓ_2 and using (4)–(5) gives $-59072 = 0$, a contradiction. Thus $v \neq 0$. If X were diffeomorphic to S^4, the induced isomorphism would send v to an element of $G(S^4, 494)$, hence to zero, contradicting injectivity. □

The Lean theorem corresponding to Theorem 3.1 accepts the input structures `ExternalGeometry` and `CSExternalGeometry`; no inhabitants of these structures are constructed in the repository. We name the candidate-specific assumptions needed to instantiate them.

Hypothesis 3.2 (Compact geometric presentation, P0)**.** The public compact Aitchison–Rubinstein product-ribbon ledger is a labelled and framed handle presentation of Σ^0_A. Its two product cancellations give the registered genus-two words, and its six-sweep tubular point-push induces the public 44- and 88-strand braids on all oriented cable states.

Hypothesis 3.3 (Coefficient comparison, C)**.** The completed MWW coefficient quantum trace admits a grading-preserving natural map to the

BPW/BHPW weight-one endpoint representation. Under this map the point-push action is the public Burau representation, and the divided cubic row descends through every beta and psi relation. Its selected value is -59072 up to an invertible convention unit, and hence is nonzero in quantum degree 494.

Hypothesis 3.4 (Three-handle closure, S)**.** The three compact owner lifts admit a complete embedded framed sphere system. For each sphere the target row composed with the MWW undotted map is zero and with the once-dotted map is the source row. Hence the divided detector descends through the complete three-handle coequalizer.

Hypothesis 3.5 (Candidate-level final identifications, P3)**.** MWW's three-/four-handle formulas identify the final quotient with $\mathcal{S}^2_{0,(0,494),0}(\Sigma^0_A;\mathbb{Q})$. The corresponding summand for S^4 is zero, and diffeomorphisms induce grading-preserving equivalences. Moreover, $\det(A-I)=1$ makes Σ^0_A a homotopy sphere. The general published theorems supporting these statements are not the candidate-specific identifications asserted here.

Theorem 3.6 (Conditional trace-73 theorem)**.** *Assume Hypotheses 3.2, 3.3, 3.4, and 3.5. Then the resulting trace-73 candidate satisfies*

$$\Sigma^0_A \simeq S^4 \quad \textit{and} \quad \Sigma^0_A \not\cong_{\mathrm{diff}} S^4.$$

Proof of Theorem 3.6. Hypotheses 3.2 and 3.5 and the Cappell–Shaneson criterion give the homotopy-sphere assertion. Hypotheses 3.3 and 3.4 instantiate the maps and rows in Theorem 3.1; its selected class is nonzero in quantum degree 494. MWW's four-handle isomorphism preserves this degree, while the same degree of the standard S^4 module vanishes. Diffeomorphism invariance and Theorem 3.1 rule out a diffeomorphism to S^4. □

4. The finite computational layer

The public finite layer has three independent parts: lattice arithmetic, a chosen-sphere coordinate determinant, and a truncated Burau calculation.

4.1. **Integral lattice calculations.** Besides (2), the repository records the proposed sphere coordinate columns

$$C = \begin{pmatrix} -1311 & -189 & 41 \\ 8608 & 1241 & -269 \\ -1 & 0 & 1 \end{pmatrix}, \qquad \det C = 1. \tag{6}$$

Thus the three recorded coordinate vectors form a basis of $\mathbb{Z}^3$. This is only the last arithmetic step in a geometric-basis argument. It does not construct embedded spheres, prove disjointness, or identify $\mathbb{Z}^3$ with $H_2^{\mathrm{sph}}(\partial W_2)$.

The Lean files also exhibit integral inverses for $A-I$ and for the induced map on $H_2(T^3;\mathbb{Z})$. These computations eliminate the candidate-specific lattice algebra from the homotopy-sphere branch. Van Kampen, Wang–Mayer–Vietoris, Poincare duality, and Hurewicz–Whitehead arguments remain supplied through a topology structure rather than formalized as four-manifold topology.

4.2. **The public Burau computation.** Let $\varepsilon = t - 1$. The script `scripts/recompute_t73_delta3.py` reads a result-free JSON input and performs the following finite operations:

(1) parse 252 primitive crossing rows and reconstruct an 11,340-letter Artin word on 44 strands;
(2) cable it to a 45,360-letter word on 88 strands;
(3) apply the unreduced Burau representation over $\mathbb{Z}[\varepsilon]/(\varepsilon^7)$;
(4) substitute $t = q^{-2}$ with $q = 1 + h$, equivalently $\varepsilon = (1 + h)^{-2} - 1$;
(5) extract the coefficient of h^3 after the specified source vector and covector are applied.

Denote the resulting scalar by δ_3^{Bur}. The independently reproduced public result is

$$\delta_3^{\mathrm{Bur}} = -59072 = (-2)^3 \cdot 7384. \tag{7}$$

The public input does not contain either decimal literal 7384 or -59072; the former is, however, stored as the constant `cubicBase` in the Lean finite layer. Lean checks only the arithmetic $(-2)^3 \cdot 7384 = -59072$ and never evaluates the 45,360-letter word.

Proposition 4.1 (Exact scope of the computation)**.** *Equation (7) is a statement about the specified truncated Burau model. The script and JSON alone do not define the MWW invariant. Under the coefficient-trace comparison of Theorem 3.3, however,*

$$\delta_3^{\mathrm{MWW}} = \delta_3^{\mathrm{Bur}} = -59072,$$

up to the harmless global convention unit normalized in that theorem.

Proof. The script defines vector and covector operations in an 88-dimensional truncated polynomial representation. The displayed MWW equality is exactly the additional content of Hypothesis 3.3; it does not follow from the finite script or from the local Burau-block calculation. □

Thus the finite recomputation and the categorical comparison have distinct roles: the former supplies the integer, while the latter makes it a four-manifold invariant.

5. Lasagna modules and the intended obstruction

We recall only the features needed for the comparison. For a smooth compact oriented four-manifold W and a framed link $L \subset \partial W$, the skein lasagna module $\mathcal{S}_0^N(W; L)$ is generated by decorated surfaces in W, modulo local relations coming from KhR_N cobordism maps [15, 14].

For a handle decomposition

$$W_1 \subset W_2 \subset W_3 \subset W_4,$$

where W_1 is a one-handlebody, W_2 adds two-handles along a framed link, W_3 adds three-handles along embedded spheres, and W_4 adds four-handles, the published results have the following roles.

- MWW Theorem 3.2 identifies the two-handle stage with a cabled skein lasagna quotient, whose relations include beta and psi operations.
- MWW Theorems 3.7 and 3.10 describe each three-handle as a coequalizer of the two hemisphere maps, and give the complete handle formula.
- MWW Proposition 3.4 says that a four-handle attachment induces an isomorphism for the empty boundary link.
- MWW Corollary 3.5 gives $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0,0)$.

At $N=2$, a nonzero class in bidegree $(0,494)$ would therefore obstruct a diffeomorphism to S^4. The intended proof constructs a functional at the one-handle stage, shows that it annihilates the two-handle and three-handle relations, and transports the resulting nonzero class through the four-handle isomorphism.

The formal algebra correctly captures the last sentence. The next sections record partial constructions and the exact candidate-specific inputs still needed by that algebra.

6. Compact product-ribbon presentation

Start with the Aitchison–Rubinstein product-annulus handle construction [1]. For the jth coordinate circle, order the integer-face crossings of the straight segment from 0 to Ae_j by their rational times. A fixed product perturbation resolves simultaneous events. This gives the mapping-torus attaching word

$$m_j = t\phi_A(x_j)t^{-1}x_j^{-1}$$

with its product-annulus normal.

The section handle cancels the base one-handle t without depositing a twist. Since $Ae_1 = e_3$, the remaining word $m_1 = zx^{-1}$ gives a second product cancellation and replaces x by z. The resulting public ledger has

$$\begin{aligned} |m_2| &= 311, \quad [m_2]_{(y,z)} = (40, 269), \\ |m_3| &= 1460, \quad [m_3]_{(y,z)} = (189, 1271). \end{aligned}$$

The word $r_{zx} = [z,z]$ formally contracts through two product bigons. If this word projection lifts to the claimed product-annulus embedding, it gives the desired framed genus-two presentation of Σ_A^0. The current generator does not output an embedded framed link or a labelled ambient isotopy/Kirby-move ledger, so this lift remains Hypothesis 3.2. This is not a merely cosmetic omission: tying a local knot into a component inside a ball preserves its free-group word and integer framing but can change the framed surgery manifold. Thus no word/framing-name argument can recover the required whole-link embedding.

The point-push collar is generated by six affine sweeps, each with 42 pure crossing factors. The generator regenerates all 252 public rows and the 11,340-letter braid. Its return wicket closes a framed loop of the r_{xy} product connector through the complement of the m_2 passages. The rows alone do not construct a tubular ambient motion in the actual handlebody. Such a motion Φ_t, together with its simultaneous action on all parallel copies, must be extracted from a solution of P0 before the public 88-strand cabling can be identified with the candidate's MWW state.

7. The coefficient-trace comparison

The compact words give

$$p_y = 42 + 2 = 44, \qquad p_z = 269 + 2 = 271.$$

The counts suggest product subrectangles pairing the 44 y wickets with 44 z passages and 227 remaining circles. The desired cut coefficient would then have the two-sided Hattori form

$$M_R(T, T') \cong \mathrm{Hom}(B_{act}T, B_{act}T')\{-44\} \otimes A^{\otimes 227}, \qquad A = \mathbb{Q}[X]/(X^2).$$

MWW Theorem 4.7 identifies the one-handle module with the coefficient trace of the cut tangle categories and describes cobordism maps componentwise [14]. BPW's functorial quantum vertical-to-horizontal trace and annular functor give

$$q\,\mathrm{Tr}(\mathcal{C}; M_R) \longrightarrow hTr_q(BN) \xrightarrow{F_{A_q}} \mathrm{gRep}(U_q(\mathfrak{sl}_2))$$

as constructed by BPW [4]. BHPW supplies strict integral tangle/foam functoriality [3]. The base change $q \mapsto 1+h$ is a rational localization followed by a Noetherian adic completion and is flat.

The endpoint type must be stated carefully. Put

$$E_{86} = (V^{\otimes 86})_{\lambda=86}, \qquad E_{88} = (V^{\otimes 88})_{\lambda=86}.$$

Then E_{86} has rank one and E_{88} has rank 88; after restriction to weight spaces the cup lies in $\mathrm{Hom}_{\mathbb{Q}[[h]]}(E_{86}, E_{88}) \cong E_{88}$. The tangle maps are $U_q(\mathfrak{sl}_2)$ intertwiners before this restriction. It is therefore incorrect to place a "weight-86 summand" on an intertwiner space itself.

There is also no abstract obstruction at the representable-coefficient step. For a rational-linear autofunctor B, the coefficient bimodule $(T, T') \mapsto \mathrm{Hom}(BT, BT')$ is two-sided equivalent to the regular coefficient bimodule, and hence their coefficient HH_0 groups are linearly equivalent. This quotient statement is kernel checked in `Smooth4PC/RepresentableCoefficient.lean`. Its use here still requires the missing candidate-specific, action-compatible Hattori equivalence from M_R to that representable model, as well as its q-deformed completed lift.

On the weight-one subspace, the normalized checked R-matrix is

$$\begin{pmatrix} 1 - q^{-2} & q^{-1} \\ q^{-1} & 0 \end{pmatrix}.$$

After conjugation by $\operatorname{diag}(1, q^{-1})$ and setting $t = q^{-2}$, this becomes the public Burau block. This is a valid local comparison. It does not construct a map from the candidate's MWW coefficient trace to the specified qHH0/Chern summand, nor the single coordinate change P that must transport the actual operator, source vector, and covector simultaneously.

The compact word has zero writhe and identity permutation. Direct evaluation of all 88 basis vectors proves

$$\rho_h(W) - I \in h^3 \operatorname{End}(E_{88}).$$

An exact noncommutative Magnus expansion of the Artin action on F_{44} shows that the automorphism is the identity through degrees one, two and three. All intermediate integer coefficients have absolute value at most four. Darné's Andreadakis equality for the pure braid group then gives

$$W \in \Gamma_3(P_{44}).$$

See [6, Theorem 6.2]. Every physical cabling is a group homomorphism and hence preserves the lower central series. Since each local pure generator acts as $I + O(h)$, every statewise cabling satisfies $\rho_h(W_s) - I = O(h^3)$. If the missing candidate-specific map and simultaneous transport exist, BPW's tangle action gives statewise equations $Q_e W_s = W_t Q_e$ for the beta and psi edges. General functoriality transports an identified action; it does not construct this missing arrow. Finally, the core identities

$$(\epsilon \otimes \epsilon)\Delta(1) = 0, \qquad (\epsilon \otimes \epsilon)\Delta(X) = 1$$

give the required Frobenius equations after the actual maps have been bound. Thus the public results prove the order-three and local algebraic parts of C, while the coefficient-trace comparison and complete beta/psi descent remain Hypothesis 3.3.

8. Three-handle sphere closure

In owner order $(m_2, m_3, r_{xy}, r_{yz}, r_{zx})$, the cellular boundary is

$$\partial_2 = \begin{pmatrix} 40 & 189 & 0 & 0 & 0 \\ 269 & 1271 & 0 & 0 & 0 \end{pmatrix}.$$

The first minor has determinant -1, so its kernel is generated by the last three owners. The unique lifts of the columns in (6) are

$$(0, 0, -1311, 8608, -1), \quad (0, 0, -189, 1241, 0), \quad (0, 0, 41, -269, 1).$$

A 32-step Nielsen ledger realizes their unimodular matrix algebraically by permutations, orientation reversals and formal sphere slides. This does not construct embedded, framed, pairwise-disjoint spheres in the actual boundary. Horvat–Jabłonowski's Theorem 5.3 additionally requires the candidate boundary, handle-count, irreducibility and spherical-basis hypotheses [10]; these have not been instantiated from a public embedded presentation.

Under Hypothesis 3.4, let Σ_j be one chosen sphere with small disk removed. MWW's raw three-handle maps attach Σ_j with zero or one dot and then the signed parallel core disks specified by its owner lift. The target detector row closes the new cable components by those same core counits. Hence the complete composite foam should be the old detector surface disjoint union an undotted or once-dotted sphere. This conclusion also requires the actual pivotal adapters, foam signs, normal-level choices and qHH0 coordinate maps; strict functoriality does not select them. Once they are supplied,

$$\mathrm{ev}(S^2) = 0, \qquad \mathrm{ev}(\dot{S}^2) = 1.$$

the displayed equations give the desired core-counit algebra. The stable movie generator explicitly records the actual MWW transport as open. Therefore the claim that all six candidate-specific sphere relations lie in the kernel remains Hypothesis 3.4.

9. Final identifications

MWW Theorem 3.10 identifies the iterated quotient with the module after the three-handles, and Proposition 3.4 says the final four-handle induces a bidegree-$(0,0)$ isomorphism [14]. If P1 and P2 have produced the specified selected class, it remains nonzero in

$$\mathcal{S}^2_{0,(0,494),0}(\Sigma^0_A; \mathbb{Q}).$$

MWW Corollary 3.5 gives $\mathcal{S}^2_0(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0,0)$, so the corresponding rational degree-494 summand vanishes. The intrinsic definition transports lasagna fillings under diffeomorphisms and gives grading-preserving equivalences. Finally, Iwaki's Proposition 2.1 and (2) prove that the genuine Cappell–Shaneson manifold Σ^0_A is a homotopy sphere [11]. Applying these general results to the proposed compact presentation and quotient requires P0–P3; the general results do not close those joins.

10. Formalization boundary

The Lean 4 development [7] is valuable precisely because it exposes the logical interface. The theorem in `Smooth4PC/T73Conditional.lean` has the form

$$\texttt{ExternalGeometry} \longrightarrow \texttt{CSExternalGeometry} \longrightarrow \big(\mathrm{HS}(X) \wedge \neg\,\mathrm{Diff}(X, S^4)\big).$$

It does not construct either input structure.

The finite files check:

- the matrix A, $A - I$, and their determinants;
- explicit integral inverse formulas for relevant lattice maps;
- the determinant of the proposed sphere coordinate matrix;
- the arithmetic value and nonvanishing of the stored cubic constant;
- the quantum degree 494 from its four recorded integer contributions;
- general linear-algebra quotient and descent lemmas.

The geometric interfaces contain, among others, the predicates

`IsActualHHO`, `IsActualSphereMap`,
`IsActualMWWCoequalizer`, `IsActualFourHandle`.

These are predicate fields of the ambient universe rather than definitions of smooth topology. The present paper does not supply their mathematical instances: P0, C, and S state missing candidate-level data. It would be incorrect to call the Lean development a formalization of four-dimensional differential topology or an unconditional proof.

The audit reports that the printed theorem dependencies use only standard logical quotient principles and classical choice, with no `sorryAx`. This is a statement about kernel checking of the algebraic core. The truth of the geometric inputs is neither established by the Lean kernel nor by the partial constructions in Sections 5–8.

11. Status table and reproducibility ledger

The table records the current allocation of every external layer. Discharged means that a public generator or primary theorem and an independently checkable argument are both supplied.

Item	Verdict	Reason
P0	Open	Word and braid ledgers do not construct the embedded framed Kirby equivalence.
P1/C	Open	Cubic order and the ordinary representable-coefficient reduction are proved, but the actual product bimodule equivalence, its completed lift and simultaneous transport are missing.
P2/E7	Open	Owner homology lifts do not construct the required embedded disjoint spheres or verify the HJ hypotheses.
P2/E10/S	Open	Core-counit algebra is proved conditionally; the actual MWW hemisphere maps and coordinate transports are missing.
P3/E11	Partial	MWW Proposition 3.4 is general; its candidate-level input depends on P1/P2.
P3/E12	Partial	The standard S^4 support is known, conditional on matching the grading and invariant.
P3/E13	Partial	Iwaki's criterion applies to Σ^0_A, conditional on P0 identifying the presentation.

The historical availability script reports the large files as absent. The compact generators provide useful partial replacements but not the embedded geometry and functorial maps listed in the table.

12. Related work

The Cappell–Shaneson literature contains both the original construction [5, 1] and substantial standardness results [2, 9, 12, 11]. Our use of Iwaki is limited to the standard matrix form, the representative table, and the homotopy-sphere criterion. His table entry for $(41, 189, 73)$ is motivation, not a singularity or exoticness theorem.

Manolescu–Neithalath and Manolescu–Walker–Wedrich provide the relevant handle calculus for skein lasagna modules [13, 14]. BPW quantum trace functoriality and the strict BHPW tangle theory provide general comparison tools [4, 3]; the candidate-specific comparison with the public endpoint coordinates remains open.

Ren and Willis use Khovanov-theoretic skein lasagna methods to distinguish other exotic four-manifolds [16]. Their computations are independent of the trace-73 construction and are not used as its comparison theorem.

Horvat–Jabłonowski's recent work gives a useful geometric criterion for recognizing three-handle attaching systems [10]. It reduces the replacement problem to boundary, embeddedness, disjointness and homology-basis checks. The compact owner-lift and Nielsen ledgers establish only the coordinate part of those checks.

13. Reproducibility and review

The proved finite and conditional layers can be replayed in the following order.

(1) Generate the compact Kirby, point-push, Hattori and owner-lift ledgers.
(2) Verify the full 88-dimensional order-three matrix statement and its mutation controls.
(3) Compile the Lean cubic, transported-pairing, Frobenius and quotient lemmas.
(4) Audit Hypotheses C and S against the cited BPW/BHPW/MWW functoriality and handle formulas.
(5) Compile the paper and compare the field-by-field allocation with `T73External.lean`.

The historical large artifacts are unavailable for this replay. Independent review should concentrate on the missing compact product-ribbon isotopy, the proposed BPW cabling comparison in Hypothesis C, and the complete-sphere closure in Hypothesis S.

14. Conclusion

The finite calculation and exact Magnus certificate provide a reproducible cubic-order signal for the public braid, and the Lean development verifies the abstract nonvanishing chain. A counterexample conclusion would follow from Theorem 3.6, but P0, C, and S remain open: the present data do not identify an embedded framed Kirby presentation, a genuine MWW coefficient functional, or the actual three-handle sphere maps. The paper therefore establishes a conditional theorem and a precise closure programme, not a disproof of the smooth four-dimensional Poincare conjecture.

Appendix A. Retired historical objects

The following table records historical artifacts cited by earlier project notes. They are unavailable; the compact partial constructions do not yet replace every load-bearing role. Their digests are retained for provenance.

Priority	Object	SHA-256
P0	`cs_presentation.py`	`B20BDFF6FBF2A0E629083E2E62DD1DDF` `9172A4E8893311CE122FFC4FEB388DFA`
P0	`t73_eps0.erkmo.json`	`6DF4D7906ADB18BA2D3237D4B49F8985` `A5B18D8C39FF522AD6498F2369803FCB`
P0	`t73_reduced_billiard.pd.json`	`E6912A64457557469E5C691B4D57ABDB` `BF4C45ADB05492777C574223D0C06F8A`
P0	`CUT_OBJECT.json`	`4E1C4564ED98F30EAEB0A1AEAAD95ACA` `B5AD7B226C79B7D5BF3F259E8B56F541`
P0	`BASE_INTERACTION_SUPPORT.json`	`2D9E4760F836D9E04E6E0024CAE03A3D` `BB49A74FF5D129756F25450B30F10F7C`
P0	`NORMAL_FIELD_MOVIE_CERT.json`	`B2E97EB0981036768B8F1C1D5EF16F8` `817EC1A7B951B6B32DBA2F9211FFD725A`
P0	`T73_COLLAR_BRAID.json`	`13AED414447ED33057C99D340910B501` `09983CF36874C14ECE0E6A2834A12C24`
P0	`verify_t73_collar_braid.py`	`9D93A6CB320200A906F461EF3131127E` `A40C80A9A7828128E6F2804808417056`
P1	`ACTUAL_PD_CABLE_UNIT_CERT.json`	`7F3D3618D6A790A9B60EE8085B647AC2` `AB742E1BC9C15841F1BEF015034217B5`
P1	`PRODUCT_NORMAL_CHRISTOFFEL_` `THXY_MOVIE.json`	`DE1AC479699EC79DE76D4265993DE437` `493A3AAA6CABB636F98998644BF3181C`
P2	`TH1_EXHAUSTIVE_ALL_ROW_GEOMETRY.json`	`EE620E6B085A5F9E1C73CFDD1AD04FC` `0682CEC74DA3DBF8AFE70DD19C038E3A0`
P2	`TH2_EXHAUSTIVE_ALL_ROW_GEOMETRY.json`	`4D1B627C0343A1C464319704EAFADCA1` `27902A2E4E90CF1C283004359B7ADC24`
P2	`THXY_FULL_MACRO_P3FREE_HJ_CERT.json`	`EABF67C0D1AE309F0281297710A283B8` `ED5A11D88C8E810837932313F4C53227`

The same values and candidate paths are machine-readable in `audit/geometric_evidence_manifest.json`. These objects are not included. Any proof of Hypotheses P0–P2 must either recover them or provide public constructions that genuinely replace their geometric and functorial roles.

Appendix B. Correspondence with the Lean fields

For readers checking the formal boundary, the principal allocation is:

Lean field	Mathematical obligation
`x0`, `ell0`, `ell0_x0`	P1 comparison and the selected genuine MWW class.
`q01`, `ell1_comp_q01`	Complete two-handle beta/psi quotient and descent.
`q12`, `ell2_comp_q12`	Actual E7 spheres, MWW hemisphere maps, and complete coequalizer.
`transport`, `fourIso`	E11 applied to the same graded lasagna module.
`s4DegreeZero`	E12 after identifying G with rational $N = 2$ lasagna.
`diffeomorphismEquiv`	Graded diffeomorphism invariance of that same object.
`matrixConditionsToHomotopySphere`	E13 for the manifold identified by P0.

These are exactly the candidate-level obligations isolated in Sections 5–8. They have not been discharged. The Lean theorem consumes no stronger statement, but it still requires them as structure fields.

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